MILITARY SPECIFICATION

FUZE, ROCKET, M412
(LESS BOOSTER PELLET)
ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-F-60005 (MU), dated 13 November 1963 and is approved by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Add the following preamble underneath the title:

"This specification is approved by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense."

PAGE 2

Add new subparagraph 3.1.1 as follows:

"3.1.1 Parts. All parts used in the assembly of the fuze shall meet all requirements of applicable drawings and specifications."

PAGE 3

3.4, Areas A, B, and C: Delete in their entirety and substitute the following:

"Area A: Area A is described as the first 0.375 inch of the firing pin hole as measured from the forward end of the base. The total porosity of all voids in this area shall not exceed 0.007 square inches (0.095 diameter).

Area B: Area B is described as a length of 0.095 inch along the length of the firing pin hole adjacent to Area A. The total porosity of all voids in the area shall not exceed 0.0007 square inch (0.030 diameter).

FSC: 1340

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:02Z
Check the source to verify that this is the current version before use.

Area C: Area C is described as the remainder of the length of the firing pin hole. The total porosity of all voids in this area shall not exceed 0.0013 square inch (0.040 diameter)."

PAGE 4

3.9.1: Add the following:

"The center of gravity is defined as a point 0.35 inches from the base end of the assembly. Functioning is defined as movement of the leaves sufficient to permit leaf number 1 to clear the rotor pin."

3.9.2:

Line 3 after "while" add: "the base plate and rotor assembly is"

Add the following sentence at the end of the paragraph: "Upon completion of the functioning time test the rotor shall be in the fully armed position."

PAGE 5

3.11.1: Add the following sentence at the end of paragraph:

"The center of gravity is defined as a point 0.50 inches from the base end of the assembly."

PAGE 7

4.2.2: Delete in its entirety and substitute the following:

"4.2.2 Inspections to be performed. Each fuze, subassembly and component shall be examined for every drawing dimension and drawing requirement applicable to the fuze, subassemblies or components. Also, the entire sample shall be examined and tested in accordance with 4.3.2, 4.3.3.1, 4.3.3.2, 4.3.3.3, 4.3.3.4, 4.3.3.5, 4.3.3.6, 4.3.3.7 and 4.3.3.8.1 as applicable (without regard to AQL or sample size specified therein). The 24 fuzes after testing in accordance with 4.3.3.8.1 shall be numbered 1 through 24 and shall be tested in accordance with the schedule in the sequence specified in 4.3.3.9."

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:02Z
Check the source to verify that this is the current version before use.

PAGE 8

4.3.2.1, Major Categories: Delete Major defect 101 in its entirety.

PAGE 13

4.3.2.12: Delete Critical defect 1 in its entirety and substitute the following:

"1. Stud leaf No. 1 not peened secure ... Visual 12001.

2. Stud leaf No. 2 not peened secure ... Visual 12008.

3. Distance between the front surface of the setback support and the rear surface of leaf number 1 above max. ... Gage 12009."

PAGE 14

4.3.2.14, Major Cagegories: Delete Major defect 101 in its entirety.

PAGE 16

4.3.2.17, Critical Categories:

Critical defect 1: Delete "(see 4.4.4)" and substitute "(see 4.3.4 and 4.4.4)."

Add new critical defect 2 as follows:

"2. One or more electrical test marks missing (tell-tale contact-cover, terminal cover) ... Visual 16008."

PAGE 17

Add new paragraph 4.3.2.20 as follows:

"4.3.2.20 Base plate and rotor assembly (see dwg. 10048786, covering a detail of dwg. 10048788).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:02Z
Check the source to verify that this is the current version before use.

Categories	Defects	Method of Inspection	Code No.
Major:	AQL 0.015 Percent		
101.	Failure to function within maximum time on functioning time test (see 3.9.2)	Test (see 4.4.11.2)	14011
Minor:	None defined."		

4.3.3.1:

Line 5: Delete "each component heat treatment lot" and substitute "each spring type included in each heat treatment batch."

Last line: Delete "lot" and substitute "batch".

4.3.3.2: Delete "Porosity of mounting base (see 3.4) - Major defect - Code No. 11004" and substitute "Porosity and surface finish of mounting base (see 3.4 and dwg. 10048784) - Porosity - Major defect - Code No. 11004, Surface Finish - Major defect - Code No. 11006."

Add new subparagraph 4.3.3.2.1 as follows:

"4.3.3.2.1 Surface finish of graze liner shaft of mounting base (see dwg. 10048784) - Major defect - Code No. 11005. A sample of 10 mounting bases shall be selected from each lot and tested as specified in 4.4.7.1. If one or more mounting bases is classed defective, the lot shall be rejected."

PAGE 18

4.3.3.6: Delete in its entirety and substitute the following:

"4.3.3.6 Functioning of graze elements (see 3.11.1) - Major defect - Code No. 14009; Examination for positioning of locking shaft (see dwg. 10048786) - Major defect - Code No. 14002. This test and examination shall be performed 100 percent on base plate and rotor assemblies (see dwg. 10048786) by assembling a firing pin, firing pin spring, and base plug to each assembly prior to assembly of the primer in the rotor. The assembly shall then be tested as

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:02Z
Check the source to verify that this is the current version before use.

specified in 4.4.10 using test equipment specified in 4.3.4. Assemblies classed defective may be disassembled and the firing pin or the firing pin spring or the locking shaft or any combination thereof, may be replaced and retested as often as necessary to obtain acceptable assemblies. Acceptable assemblies shall be reassembled and visual examination performed under three power magnification to assure that the locking shaft is bearing properly on the graze lever. Assemblies which contain locking shafts which do not bear on the graze lever, as illustrated on the applicable drawing, shall be classed defective and removed from the lot, subject to repositioning of the locking shaft and re-examination prior to re-entry into the lot. (If the graze elements are shipped separately, acceptable assemblies shall be reassembled to yield base plate and rotor assemblies and a set of graze elements consisting of the firing pin and firing pin spring. The set of graze elements shall be identical so that subsequent assembly of the graze elements into the proper base plate and rotor assemblies is assured.)."

4.3.3.7:

Lines 1 to 4: Delete in their entirety and substitute the following:

"4.3.3.7 Operation of setback assembly (see 3.9); functioning during non-functioning test-Critical defect - Code No. 14010. Thes test shall be performed ..."

Line 6: Delete "4.4.11" and substitute "4.4.11.1".

4.3.3.8.1: Delete in its entirety and substitute the following:

"4.3.3.8.1 At plus 70 degrees Fahrenheit (F.) (see 3.10.1) - Arming time, minimum - Critical defect - Code No. 14012; and Arming time, maximum - Major defect - Code No. 14013. This test shall be performed 100 percent on base plate and rotor assemblies conditioned within the temperature limitations of 65°F. to 85°F., and the assemblies shall comply with the requirements as specified in paragraph 3.10.1. The test shall be performed as specified in 4.4.12 using test equipment specified in 4.3.4. Assemblies failing to arm in the time specified shall be classed defective and removed from the lot."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:02Z
Check the source to verify that this is the current version before use.

4.3.3.8.2: Delete in its entirety.

4.3.3.9.2: Delete in its entirety and substitute the following:

"4.3.3.9.2 Rough handling (see 3.8) - Functioning during non-functioning test (see 3.9.1) - Critical defect - Code No. 16009; Functioning time, maximum (see 3.9.2) - Major defect - Code No. 16010; Rotor arming time at plus 70°F., minimum (see 3.10.1) - Critical defect - Code No. 16011; Rotor arming time at minus 40°F., minimum (see 3.10.2) - Critical defect - Code No. 14014; Rotor arming time at plus 70°F., maximum (see 3.10.1) - Major defect - Code No. 16012; and Rotor arming time at minus 40°F., maximum (see 3.10.2) - Major defect - Code No. 14015. The remaining 18 fuzes shall be subjected to the transportation vibration test specified in Standard MIL-STD-353 and then without disassembly to the five foot drop test specified in Standard MIL-STD-358 except that the fuze shall be dropped bare, once only, and impact shall be on the base. The cover assembly, graze elements, and base plug components shall be removed but the graze elements (firing pin and firing pin spring) identified for later re-insertion into the same fuze and the fuzes shall be subjected to the following tests in the sequence specified using test equipment specified in 4.3.4:

Test	Test Procedure
Non-functioning of setback assembly	4.4.11
Functioning of setback assembly	4.4.11
Rotor arming time at plus 70°F.	4.4.12
Rotor arming time at minus 40°F.	4.4.12

The temperature tolerance adhered to during performance of the rotor arming time test at plus 70°F. shall be plus 10°F. and minus 0°F. and at minus 40°F. shall be plus and minus 5°F. The lot shall be rejected if one or more fuzes fail to meet any of the applicable requirements, or if one or more setback assemblies function during the non-functioning test, or if the setback functioning time fails to conform to the specified requirements, or if one or more rotor assemblies arm in less than the minimum time specified at either

Source: https://assist.dla.mil -- Downloaded: 202640-23T00:02Z
Check the source to verify that this is the currentversion before use.

conditioning temperature, or if the average rotor arming time plus the product of 1.6 and the standard deviation exceeds the maximum time specified at either conditioning temperature, or if the average rotor arming time minus the product of 1.6 and the standard deviation is less than the minimum time specified at either conditioning temperature. Outliers may be eliminated at the upper limit only (see 6.5). The cover assembly, graze elements, and base plugs shall be reassembled to the fuzes, and the fuzes tested as follows in 4.3.3.9.3 and 4.3.3.9.4."

PAGE 20

4.3.3.9.3:

Line 2: Delete "14015" and substitute "16013".

Line 3: Delete "14016" and substitute "16014".

Line 9: Delete "within the g limits specified" and substitute "and fail to perforate the booster cup".

4.3.3.9.4, line 2: Delete "14017" and substitute "16015".

PAGE 22

4.4.3: Delete in its entirety and substitute the following:

"4.4.3 Dielectric strength. The specified electrical potential shall be applied between the points to be tested. Breakdown shall be defined as a leakage current in excess of 10 microamperes for the rotor assembly test and in excess of 20 microamperes for the cover assembly test. The test equipment shall be capable of detecting the defined breakdown current and shall automatically indicate a defective assembly."

4.4.4: Delete in its entirety and substitute the following:

"4.4.4 Fuze safety. The electrical resistance shall be measured between the terminal and the cover and between the telltale contact and the cover. A black dot shall be readily visible on the cover plate to indicate the fuze has passed the terminal-cover electrical resistance test. A red dot shall be readily visible on the cover plate to indicate that the fuze has passed the telltale contact-cover electrical resistance test. The test equipment shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:02Z
Check the source to verify that this is the current version before use.

separately stamp the proper marking on the fuze, only for each electrical resistance measurement which is within the required limits The fuze shall be classed defective and removed from the lot if after having been subjected to the test one or more acceptance marks are not present on the fuze.

WARNING: When measuring the resistance between the terminal and the cover, the test equipment shall limit the current to 20 microamperes to prevent inadvertent functioning of the detonator. This test may be dangerous and should be performed behind a suitable barricade."

PAGE 23

4.4.7, last two lines: Delete in their entirety and substitute the following:

"examined for porosity and surface finish. The mounting base shall be classed defective if porosity is observed in excess of the specified requirements or if the surface finish exceeds the specified requirements."

Add new subparagraph 4.4.7.1 as follows:

"4.4.7.1 Surface finish of graze lever shaft of mounting base. The mounting base shall be sectioned through the center of the graze lever shaft and each of the sections shall be examined for surface finish using the equipment specified in 4.3.4. The mounting base shall be classed defective if it fails to comply with the specified surface finish requirement."

4.4.11: Delete in its entirety and substitute the following:

"4.4.11 Operation of setback assembly.

4.4.11.1 Non-functioning. The assembly shall be mounted on a horizontal centrifuge with the longitudinal axis of the assembly lying along a path of radial acceleration, and the base end pointed away from the center of rotation. The minimum non-arming acceleration shall be applied for the minimum time specified and observation made for functioning or non-functioning. The assembly shall be classed defective if the setback assembly functions during the non-functioning phase.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:02Z
Check the source to verify that this is the current version before use.

4.4.11.2 Functioning time. The assembly shall be mounted as in 4.4.11.1. The test equipment shall lock leaf number 3 in the fully safe position and the required acceleration applied to the assembly and stabilized. The number 3 leaf shall be released and the time interval measured from release of leaf number 3 until leaf number 1 moves to a position that would release the rotor. The assembly shall be classed defective if the measured time interval exceeds the specified requirement or if the rotor fails to be in the fully armed position after the arming acceleration has been removed."

PAGE 24

4.4.12: Delete first sentence and substitute the following:

"4.4.12 Rotor arming time. The assemblies shall be conditioned within the specified temperatures for a minimum of three (3) hours."

4.4.13, lines 5 and 6: Delete "or until the maximum functioning graze acceleration is attained."

PAGE 25

5.1.2.2: Add the following sentence:

"If the fuzes are shipped with the graze elements not assembled, the base plugs shall be packaged, packed, and marked in accordance with Specification MIL-P-10025."

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project No: 1340-A606

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:02Z
Check the source to verify that this is the current version before use.